# Summary


- Every idempotent matrix $\boldsymbol{P}^2 = \boldsymbol{P}$ represents a projection. If, in addition, the matrix is self-adjoint $\langle \boldsymbol{Px}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{Py} \rangle$, then the projection is orthogonal.
- If $V$ is expressible as the direct sum of two subspaces $V_1 \oplus V_2$, then the projection onto $V$ is the sum of the projections onto $V_1$ and onto $V_2$.
- The orthogonal projection $\hat{\boldsymbol{u}}$ of $\boldsymbol{u}$ onto a subspace $V$ has two defining properties: (i.) $\hat{\boldsymbol{u}} \in V$ and (ii.) $\boldsymbol{u} - \hat{\boldsymbol{u}} \in V^{\perp}$.
- Every finite dimensional subspace $V$ of a Hilbert space $\mathcal{H}$ is closed.
- If $V$ is a closed subspace of $\mathcal{H}$, then for any $\boldsymbol{u} \in \mathcal{H}$, there exists a unique closest element in $V$ to $\boldsymbol{u}$, and that element is the orthogonal projection $\hat{\boldsymbol{u}}$.


# Direct Sums

**Definition 2.1.** Suppose $V_1$ and $V_2$ are linear subspaces. The sum $V = V_1 + V_2$ consists of those vectors $\{\boldsymbol{v}_1 + \boldsymbol{v}_2 : \boldsymbol{v}_1 \in V_1, \boldsymbol{v}_2 \in V_2\}$. If $V_1$ and $V_2$ are linearly independent, then $V$ is described as the **direct sum** of $V_1$ and $V_2$, written $V = V_1 \oplus V_2$. ■

**Proposition 2.1.** If $V = V_1 \oplus V_2$, then any $\boldsymbol{v} \in V$ has a unique representation $\boldsymbol{v} = \boldsymbol{v}_1 + \boldsymbol{v}_2$ where $\boldsymbol{v}_1 \in V_1$ and $\boldsymbol{v}_2 \in V_2$. ♦

**Proof.** Since $V$ is the sum of $V_1$ and $V_2$, there exist $\boldsymbol{v}_1 \in V_1$ and $\boldsymbol{v}_2 \in V_2$ such that $\boldsymbol{v} = \boldsymbol{v}_1 + \boldsymbol{v}_2$. Suppose that there exists another representation $\boldsymbol{v} = \boldsymbol{u}_1 + \boldsymbol{u}_2$, then:

$$\boldsymbol{0} = \boldsymbol{v} - \boldsymbol{v} = \big(\boldsymbol{v}_1 - \boldsymbol{u}_1\big) + \big(\boldsymbol{v}_2 - \boldsymbol{u}_2\big).$$

But $\boldsymbol{v}_1 - \boldsymbol{u}_1 \in V_1$ and $\boldsymbol{v}_2 - \boldsymbol{u}_2 \in V_2$. By linear independence of $V_1$ and $V_2$, $\boldsymbol{v}_1 - \boldsymbol{u}_1 = \boldsymbol{0}$ and $\boldsymbol{v}_2 - \boldsymbol{u}_2 = \boldsymbol{0}$. Conclude the representation of $\boldsymbol{v}$ is unique. ■

**Definition 2.2.** A **basis** is a collection $B$ of *linearly independent* vectors that *span* a linear space $V$. ■

**Proposition 2.2.** Suppose $B_1$ is a basis for $V_1$ and $B_2$ is a basis for $V_2$, then the union $B = B_1 \cup B_2$ is a basis for $V = V_1 \oplus V_2$. ♦

**Proof.** Any $\boldsymbol{v} \in V$ is uniquely expressible as $\boldsymbol{v}_1 + \boldsymbol{v}_2$, where $\boldsymbol{v}_1 \in V_1$ and $\boldsymbol{v}_2 \in V_2$. Since $B_1$ and $B_2$ are bases for $V_1$ and $V_2$, $\boldsymbol{v}_1$ has a unique representation with respect to $B_1$, and $\boldsymbol{v}_2$ has a unique representation with respect to $B_2$. Thus $\boldsymbol{v}$ is in the span of $B_1 \cup B_2$. Moreover, since $V_1$ and $V_2$ are linearly independent, and $B_1$ and $B_2$ are each bases, the collection of vectors $B_1 \cup B_2$ is linearly independent. ■

**Corollary 2.1.** If $V = V_1 \oplus V_2$, then $\dim(V) = \dim(V_1) + \dim(V_2)$. ♣

# Projections

**Definition 3.1.** Suppose $V = V_1 \oplus V_2$, then every $\boldsymbol{v} \in V$ has a uniform representation $\boldsymbol{v} = \boldsymbol{v}_1 + \boldsymbol{v}_2$ where $\boldsymbol{v}_1 \in V_1$ and $\boldsymbol{v}_2 \in V_2$. The mapping $\Pi(\boldsymbol{v}|V_1) = \boldsymbol{v}_1$ is the **projection** of $\boldsymbol{v}$ onto $\boldsymbol{V}_1$. Likewise, $\Pi(\boldsymbol{v}|V_2) = \boldsymbol{v}_2$ is the projection of $\boldsymbol{v}$ onto $V_2$. ■

**Proposition 3.1.** Suppose $V = V_1 \oplus V_2$, and $\boldsymbol{v} \in V$. The projection $\Pi(\boldsymbol{v}|V_1)$ of $\boldsymbol{v}$ onto $V_1$ is a linear mapping. ♦

**Proof.** Suppose $\boldsymbol{u}$ and $\boldsymbol{v}$ are in $V$, then $\boldsymbol{u} = \boldsymbol{u}_1 + \boldsymbol{u}_2$ and $\boldsymbol{v} = \boldsymbol{v}_1 + \boldsymbol{v}_2$, with $\boldsymbol{u}_k, \boldsymbol{v}_k \in V_k$. Consider the sum $\boldsymbol{w} \equiv \boldsymbol{u} + \boldsymbol{v} = \big(\boldsymbol{u}_1 + \boldsymbol{v}_1\big) + \big(\boldsymbol{u}_2 + \boldsymbol{v}_2\big)$. Observe that $\boldsymbol{w}_k = \boldsymbol{u}_k + \boldsymbol{v}_k \in V_k$. Since this representation is unique:

$$\Pi(\boldsymbol{u} + \boldsymbol{v}|V_1) = \Pi(\boldsymbol{w}|V_1) = \boldsymbol{w}_1 = \boldsymbol{u}_1 + \boldsymbol{v}_1 = \Pi(\boldsymbol{u}|V_1) + \Pi(\boldsymbol{v}|V_1).$$

■

**Corollary 3.1.** $\Pi(\cdot|V_1)$ is expressible as a *projection matrix* $\boldsymbol{P}$. ♣

**Proposition 3.2.** Suppose $V = V_1 \oplus V_2$, and $\boldsymbol{v} \in V$. The projection $\Pi(\boldsymbol{v}|V_1)$ mapping is **idempotent**. Consequently, $\boldsymbol{P}^2 = \boldsymbol{P}$. ♦

**Proof.** The projection of $\boldsymbol{v}$ onto $V_1$ is:

$$\Pi(\boldsymbol{v}|V_1) = \boldsymbol{v}_1 = \boldsymbol{P}\boldsymbol{v}.$$

Since $v_1$ already belongs to $V_1$, upon projecting again:

$$\boldsymbol{P}\boldsymbol{P}\boldsymbol{v} = \boldsymbol{P}\boldsymbol{v}_1 = \Pi(\boldsymbol{v}_1|V_1) = \boldsymbol{v}_1 = \boldsymbol{P}\boldsymbol{v}.$$

Since $\boldsymbol{P}^2\boldsymbol{v} = \boldsymbol{P}\boldsymbol{v}$ holds for $\forall \boldsymbol{v} \in V$, conclude that $\boldsymbol{P}^2 = \boldsymbol{P}$. ■

**Proposition 3.3.** Suppose $V = V_1 \oplus V_2$, and $\boldsymbol{v} \in V$. Let $\boldsymbol{P}$ denote projection onto $\boldsymbol{V}_1$. Then, $\boldsymbol{Q} = \boldsymbol{I} - \boldsymbol{P}$ is projection onto $V_2$ and $\boldsymbol{P}\boldsymbol{Q} = \boldsymbol{Q}\boldsymbol{P} = \boldsymbol{0}$. ♦

**Proof.** The projection onto $V_2$ is:

$$\Pi(\boldsymbol{v}|V_2) = \boldsymbol{v}_2 = \boldsymbol{v} - \boldsymbol{v}_1 = \boldsymbol{v} - \Pi(\boldsymbol{v}|V_1) = \boldsymbol{v} - \boldsymbol{P}\boldsymbol{v} = \big(\boldsymbol{I} - \boldsymbol{P}\big)\boldsymbol{v}.$$

Let $\boldsymbol{Q} = \boldsymbol{I} - \boldsymbol{P}$, such that $\boldsymbol{Q}\boldsymbol{v} = \boldsymbol{v} - \boldsymbol{P}\boldsymbol{v} = \boldsymbol{v} - \boldsymbol{v}_1 = \boldsymbol{v}_2 = \Pi(\boldsymbol{v}|V_2)$. By direct calculation:

$$\boldsymbol{P}\boldsymbol{Q} = \boldsymbol{P}\big(\boldsymbol{I} - \boldsymbol{P}\big) = \boldsymbol{P} - \boldsymbol{P}^2 = \boldsymbol{P} - \boldsymbol{P} = \boldsymbol{0}.$$

Likewise, $\boldsymbol{Q}\boldsymbol{P} = \big(\boldsymbol{I} - \boldsymbol{P}\big)\boldsymbol{P} = \boldsymbol{P} - \boldsymbol{P}^2 = \boldsymbol{P} - \boldsymbol{P} = \boldsymbol{0}$. Thus:

$$\Pi\big\{\Pi(\boldsymbol{v}|V_1)|V_2\big\} = \boldsymbol{0} = \Pi\big\{\Pi(\boldsymbol{v}|V_2)|V_1\big\}$$

when $V_1$ is linearly independent of $\boldsymbol{V}_2$, or equivalently when $V_1 \cap V_2 = \{\boldsymbol{0}\}$. ■

**Proposition 3.4.** Suppose $\boldsymbol{v} \in V$ and $\boldsymbol{P}^2 = \boldsymbol{P}$, then $V = V_1 \oplus V_2$ where $V_1 = \text{im}(\boldsymbol{P})$ and $V_2 = \text{im}(\boldsymbol{I} - \boldsymbol{P})$. ♦

**Proof.** Any $\boldsymbol{v}$ in $V$ is expressible as:

$$\boldsymbol{v} = \big(\boldsymbol{I} - \boldsymbol{P} + \boldsymbol{P}\big)\boldsymbol{v} = \big(\boldsymbol{I} - \boldsymbol{P}\big)\boldsymbol{v} + \boldsymbol{P}\boldsymbol{v}.$$

Thus $V = V_2 + V_1$. To verify that $V$ is the direct sum, it suffices to show $V_1 \cap V_2 = \{\boldsymbol{0}\}$. Suppose $\boldsymbol{u} \in V_1 \cap V_2$. Since $\boldsymbol{u} \in V_2$, there exists a coefficient vector $\boldsymbol{\alpha}$ such that $\boldsymbol{u} = (\boldsymbol{I} - \boldsymbol{P})\boldsymbol{\alpha}$. Moreover, since $\boldsymbol{u} \in V_1$, $\boldsymbol{u} = \boldsymbol{P}\boldsymbol{u}$. Now:

$$\boldsymbol{u} = \boldsymbol{P}\boldsymbol{u} = \boldsymbol{P}(\boldsymbol{I} - \boldsymbol{P})\boldsymbol{\alpha} = (\boldsymbol{P} - \boldsymbol{P}^2)\boldsymbol{\alpha} = (\boldsymbol{P} - \boldsymbol{P})\boldsymbol{\alpha} = \boldsymbol{0}.$$

■

**Proposition 3.5.** All eigenvalues of a projection matrix are either 0 or 1. ♦

**Proof.** Suppose $\boldsymbol{P}$ is a projection matrix with eigenvector $\boldsymbol{u}$, then $\boldsymbol{P}\boldsymbol{u} = \lambda\boldsymbol{u}$ for some eigenvalue $\lambda$. Left multiplying by $\boldsymbol{P}$:

$$\boldsymbol{P}^2\boldsymbol{u} = \boldsymbol{P}\lambda\boldsymbol{u} = \lambda\boldsymbol{P}\boldsymbol{u} = \lambda^2\boldsymbol{u}.$$

Yet by idempotency of $\boldsymbol{P}$, $\boldsymbol{P}^2\boldsymbol{u} = \boldsymbol{P}\boldsymbol{u} = \lambda\boldsymbol{u}$. Thus $\lambda\boldsymbol{u} = \lambda^2\boldsymbol{u}$, so $\lambda \in \{0, 1\}$. ■

# Hilbert Space

**Definition 4.1 (Hilbert Space).** A **Hilbert** space $\mathcal{H}$ is a *complete linear space* with an *inner product norm*:

$$||\boldsymbol{v}||^2 = \langle \boldsymbol{v}, \boldsymbol{v} \rangle, \text{ for } \boldsymbol{v} \in \mathcal{H}.$$

*Euclidean space* is a finite dimensional Hilbert space. ■

## 4.1 Orthogonal Complement

**Definition 4.2.** The **orthogonal complement** $V^{\perp}$ to a subspace $V$ consists of those vectors orthogonal to each element of $V$:

$$V^{\perp} = \big\{\boldsymbol{u} \in \boldsymbol{\Omega} : \langle \boldsymbol{u}, \boldsymbol{v} \rangle = 0 \text{ for } \forall \boldsymbol{v} \in V\big\}.$$

■

**Proposition 4.1.** The orthogonal complement $V^{\perp}$ is (always) a closed subspace. ♦

**Proof.** Suppose $\boldsymbol{v} \in V$. If $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ are two vectors in $V^{\perp}$, then any linear combination $\boldsymbol{u} = \alpha_1 \boldsymbol{u}_1 + \alpha_2 \boldsymbol{u}_2$ remains in $V^{\perp}$ since:

$$\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \alpha_1 \langle \boldsymbol{u}_1, \boldsymbol{v} \rangle + \alpha_2 \langle \boldsymbol{u}_2, \boldsymbol{v} \rangle = 0.$$

Suppose $(\boldsymbol{u}_n)$ is a Cauchy sequence in $V^{\perp}$, then $(\boldsymbol{u}_n)$ converges to some limit $\boldsymbol{u}$. By continuity of the inner product:

$$\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \lim_{n \to \infty} \langle \boldsymbol{u}_n, \boldsymbol{v} \rangle = \boldsymbol{0}.$$

■

**Proposition 4.2.** The following inclusions hold:

i. $V \subseteq V^{\perp\perp}$.

ii. If $U \subseteq V$, then $V^{\perp} \subseteq U^{\perp}$.

♦

**Proof.** (i.) If $\boldsymbol{v} \in V$, then $\langle \boldsymbol{v}, \boldsymbol{u} \rangle = 0$ for $\forall \boldsymbol{u} \in V^{\perp}$, therefore $\boldsymbol{v} \in V^{\perp\perp}$.

(ii.) If $\boldsymbol{v}_{\perp} \in V^{\perp}$, then $\boldsymbol{v}_{\perp}$ is orthogonal to each $\boldsymbol{v}$ in $V$. Now $U \subseteq V$, so $\boldsymbol{v}_{\perp}$ is orthogonal to each $\boldsymbol{u} \in U$. Conclude $\boldsymbol{v}_{\perp} \in U^{\perp}$. ■

## 4.2 Finite Closure

**Theorem 4.1.** Suppose $V$ is a *finite* subspace of a Hilbert space $\mathcal{H}$, then $V$ is complete, and therefore closed. □

**Proof.** Suppose $V$ is a finite dimensional subspace with basis $\boldsymbol{B} = (\boldsymbol{b}_1, \cdots, \boldsymbol{b}_n)$, then every $\boldsymbol{v} \in V$ is expressible as $\boldsymbol{v} = \alpha_1 \boldsymbol{b}_1 + \cdots + \alpha_n \boldsymbol{b}_n$ for some coefficients $(\alpha_1, \cdots, \alpha_n)$. There exists a bijection $T : V \to \mathbb{F}^n$ such that $T(\boldsymbol{v}) = (\alpha_1, \cdots, \alpha_n)$, where $\boldsymbol{F}$ is a complete field. Moreover, this mapping is linear since:

$$T(\boldsymbol{v}_1 + \boldsymbol{v}_2) = (\alpha_1 + \beta_1, \cdots, \alpha_n + \beta_n) = (\alpha_1, \cdots, \alpha_n) + (\beta_1, \cdots, \beta_n) = T(\boldsymbol{v}_1) + T(\boldsymbol{v}_2).$$

Suppose $(\boldsymbol{v}_n)$ is a Cauchy sequence in $V$. For $\epsilon > 0$, define $\delta = \epsilon / ||T||$, where:

$$||T|| = \sup_{\{\boldsymbol{v} \neq \boldsymbol{0}\}} \frac{||T(\boldsymbol{x})||}{||\boldsymbol{x}||}.$$

Since $\boldsymbol{v}_n$ is Cauchy, there exist a threshold $\nu$ such that when $n, m \geq \nu$:

$$||\boldsymbol{v}_n - \boldsymbol{v}_m|| < \frac{\epsilon}{||T||}.$$

Now for $n, m \geq \nu$:

$$||T(\boldsymbol{v}_n) - T(\boldsymbol{v}_m)|| = ||T(\boldsymbol{v}_n - \boldsymbol{v}_m)|| \leq ||T|| \cdot ||\boldsymbol{v}_n - \boldsymbol{v}_m|| = \epsilon.$$

Conclude that $\{T_n \equiv T(\boldsymbol{v}_n)\}$ is Cauchy. Since $\mathbb{F}^n$ is complete, $(T_n)$ converges to some limit $T_0 \in \mathbb{F}^n$, and:

$$\lim_{n\to\infty} ||T(\boldsymbol{v}_n) - T_0|| = 0.$$

Since $T$ is a bijection, there exists $\boldsymbol{v}_0 \in V$ such that $T(\boldsymbol{v}_0) = T_0$. Now:

$$0 \leq \lim_{n\to\infty} ||\boldsymbol{v}_n - \boldsymbol{v}_0|| = \lim_{n\to\infty} ||T^{-1}T(\boldsymbol{v}_n) - T^{-1}T_0|| \leq ||T^{-1}|| \lim_{n\to\infty} ||T(\boldsymbol{v}_n) - T_0|| = 0.$$

Since $(\boldsymbol{v}_n)$ converges to a limit in $V$, conclude that $V$ is complete. ■

# Orthogonal Projection

**Definition 5.1.** Let $V$ denote a closed subspace of a Hilbert space $\mathcal{H}$. The **orthogonal projection** of $\boldsymbol{u} \in \mathcal{H}$ onto $V$ is the element $\Pi_0(\boldsymbol{u}|V)$ with these properties:

i. the projection resides in $V$: $\Pi_0(\boldsymbol{u}|V) \in V$.

ii. the residual is orthogonal to $V$: $\boldsymbol{u} - \Pi_0(\boldsymbol{u}|V) \in V^{\perp}$.

■

**Remark 5.1.** The notation $\Pi_0$ distinguishes the orthogonal projection from the potentially oblique projection $\Pi$. ♦

**Lemma 5.1.**

$$||\boldsymbol{x} + \boldsymbol{y}||^2 + ||\boldsymbol{x} - \boldsymbol{y}||^2 = 2||\boldsymbol{x}||^2 + 2||\boldsymbol{y}||^2. \tag{5.1}$$

Proof is by direct expansion. ■

**Theorem 5.1 (Projection Theorem).** Suppose $V$ is a closed subspace of a Hilbert space $\mathcal{H}$. For any $\boldsymbol{u} \in \mathcal{H}$, there *exists* a *unique* closest element in $V$ to $\boldsymbol{u}$:

$$\hat{\boldsymbol{u}} = \arg\min_{\boldsymbol{v}\in V} ||\boldsymbol{v} - \boldsymbol{u}||^2,$$

and $\hat{\boldsymbol{u}}$ is the *orthogonal projection* of $\boldsymbol{u}$ onto $V$. □

**Proof.** (Existence) If $\boldsymbol{u} \in V$, then $\hat{\boldsymbol{u}} = \boldsymbol{u}$. Suppose not. Let $\delta^2 = \inf_{\boldsymbol{v} \in V} ||\boldsymbol{v} - \boldsymbol{u}||^2$. Construct a sequence $\boldsymbol{v}_n \in V$ such that:

$$||\boldsymbol{v}_n - \boldsymbol{u}||^2 \leq \delta^2 + \frac{1}{n}.$$

By (**??**):

$$||(\boldsymbol{v}_n - \boldsymbol{u}) + (\boldsymbol{u} - \boldsymbol{v}_m)||^2 + ||(\boldsymbol{v}_n - \boldsymbol{u}) - (\boldsymbol{u} - \boldsymbol{v}_m)||^2 = 2||\boldsymbol{v}_n - \boldsymbol{u}||^2 + 2||\boldsymbol{u} - \boldsymbol{v}_m||^2.$$

Upon rearranging:

$$||\boldsymbol{v}_n - \boldsymbol{v}_m||^2 + 4\left|\left|\frac{\boldsymbol{v}_n + \boldsymbol{v}_m}{2} - \boldsymbol{u}\right|\right|^2 = 2||\boldsymbol{v}_n - \boldsymbol{u}||^2 + 2||\boldsymbol{u} - \boldsymbol{v}_m||^2,$$
$$||\boldsymbol{v}_n - \boldsymbol{v}_m||^2 = 2||\boldsymbol{v}_n - \boldsymbol{u}||^2 + 2||\boldsymbol{u} - \boldsymbol{v}_m||^2 - 4\left|\left|\frac{\boldsymbol{v}_n + \boldsymbol{v}_m}{2} - \boldsymbol{u}\right|\right|^2.$$

Now, for all $(n, m) \in \mathbb{N}^2$, the midpoint $\boldsymbol{\mu}_{nm} = (\boldsymbol{v}_n + \boldsymbol{v}_m)/2$ is in $V$ since $V$ is convex. Therefore, $||\boldsymbol{\mu}_{nm} - \boldsymbol{u}||^2 \geq \delta^2$, and:

$$||\boldsymbol{v}_n - \boldsymbol{v}_m||^2 \leq 2||\boldsymbol{v}_n - \boldsymbol{u}||^2 + 2||\boldsymbol{v}_m - \boldsymbol{u}||^2 - 4\delta^2.$$

By construction $\lim_{n\to\infty} ||\boldsymbol{v}_n - \boldsymbol{u}||^2 = \delta^2$, thus:

$$\lim_{(n,m)\to\infty} ||\boldsymbol{v}_n - \boldsymbol{v}_m||^2 \leq 2\delta^2 + 2\delta^2 - 4\delta^2 = 0.$$

Conclude that $(\boldsymbol{v}_n)$ is a Cauchy sequence. Since $V$ is a closed subspace of a Hilbert space, $(\boldsymbol{v}_n)$ converges to a limit in $V$. Call this limit $\hat{\boldsymbol{u}}$. By continuity of the norm:

$$||\hat{\boldsymbol{u}} - \boldsymbol{u}||^2 = \lim_{n\to\infty} ||\boldsymbol{v}_n - \boldsymbol{u}||^2 \leq \lim_{n\to\infty} \left(\delta^2 + \frac{1}{n}\right) = \delta^2.$$

Moreover since $\hat{\boldsymbol{u}} \in V$, $||\hat{\boldsymbol{u}} - \boldsymbol{u}||^2 \geq \delta^2$ by definition of the infimum. Conclude that $||\hat{\boldsymbol{u}} - \boldsymbol{u}||^2 = \delta^2$, and therefore $\hat{\boldsymbol{u}} = \arg\min_{\boldsymbol{v}\in V} ||\boldsymbol{v} - \boldsymbol{u}||^2$.

(Orthogonal projection) Since $\hat{\boldsymbol{u}}$ is in $V$, to establish $\hat{\boldsymbol{u}}$ is the orthogonal projection of $\boldsymbol{u}$ onto $V$, it remains to show that $\boldsymbol{u} - \hat{\boldsymbol{u}}$ is in $V^{\perp}$. Suppose to the contrary that there exists $\boldsymbol{v}$ in $V$ not orthogonal to $\boldsymbol{u} - \hat{\boldsymbol{u}}$. wlog let $||\boldsymbol{v}||^2 = 1$, and set $\theta = \langle \boldsymbol{u} - \hat{\boldsymbol{u}}, \boldsymbol{v} \rangle$. Define $\tilde{\boldsymbol{u}} = \theta\boldsymbol{v} + \hat{\boldsymbol{u}}$, which is in $V$. Then:

$$\begin{aligned}||\tilde{\boldsymbol{u}} - \boldsymbol{u}||^2 &= ||\theta\boldsymbol{v} + \hat{\boldsymbol{u}} - \boldsymbol{u}||^2 = ||\theta\boldsymbol{u}||^2 + ||\hat{\boldsymbol{u}} - \boldsymbol{u}||^2 + 2\theta\langle \boldsymbol{v}, \hat{\boldsymbol{u}} - \boldsymbol{u} \rangle \\ &= \theta^2 + ||\hat{\boldsymbol{u}} - \boldsymbol{u}||^2 - 2\theta^2 = ||\hat{\boldsymbol{u}} - \boldsymbol{u}||^2 - \theta^2 \leq ||\hat{\boldsymbol{u}} - \boldsymbol{u}||^2.\end{aligned}$$

Thus, if $\boldsymbol{u} - \hat{\boldsymbol{u}}$ is not in $V^{\perp}$, then $\hat{\boldsymbol{u}}$ is not the closest point in $V$ to $\boldsymbol{u}$. Since $\hat{\boldsymbol{u}} \in V$ and $\boldsymbol{u} - \hat{\boldsymbol{u}} \in V^{\perp}$ if $\hat{\boldsymbol{u}} = \arg\min_{\boldsymbol{v}\in V} ||\boldsymbol{v} - \boldsymbol{u}||^2$, conclude that $\hat{\boldsymbol{u}} = \Pi_0(\boldsymbol{u}|V)$.

(Uniqueness) Suppose $\hat{\boldsymbol{u}}_1$ and $\hat{\boldsymbol{u}}_2$ are both orthogonal projections of $\boldsymbol{u}$ onto $V$, then $\boldsymbol{u} - \hat{\boldsymbol{u}}_1 \in V^{\perp}$ and $\boldsymbol{u} - \hat{\boldsymbol{u}}_2 \in V^{\perp}$. Thus $\big(\boldsymbol{u} - \hat{\boldsymbol{u}}_1\big) - \big(\boldsymbol{u} - \hat{\boldsymbol{u}}_2\big) = \hat{\boldsymbol{u}}_2 - \hat{\boldsymbol{u}}_1 \in V^{\perp}$. But $\hat{\boldsymbol{u}}_2 \in V$ and $\hat{\boldsymbol{u}}_1 \in V$, so $\hat{\boldsymbol{u}}_2 - \hat{\boldsymbol{u}}_1 \in V \cap V^{\perp} = \{\boldsymbol{0}\}$. Conclude $\hat{\boldsymbol{u}}_1 = \hat{\boldsymbol{u}}_2$. ■

**Corollary 5.1.** If $V$ is a closed linear subspace of a Hilbert space $\mathcal{H}$, then:

$$\mathcal{H} = V \oplus V^{\perp}.$$

♣

**Proposition 5.1.** Suppose $V$ is a closed subspace of Hilbert space $\mathcal{H}$ and that $V$ is the direct sum of two orthogonal subspaces: $V = V_0 \oplus V_1$ with $V_0 \perp V_1$. Then, for any $\boldsymbol{u} \in \mathcal{H}$, the projection onto $V$ is the sum of the projections onto $V_0$ and onto $V_1$:

$$\Pi_0(\boldsymbol{u}|V) = \Pi_0(\boldsymbol{u}|V_0) + \Pi_0(\boldsymbol{u}|V_1).$$

♦

**Proof.** Let $\hat{\boldsymbol{u}}_0$ and $\hat{\boldsymbol{u}}_1$ denote orthogonal projection onto $V_0$ and $V_1$. Consider the sum $\hat{\boldsymbol{u}} = \hat{\boldsymbol{u}}_0 + \hat{\boldsymbol{u}}_1$. Since $V = V_0 \oplus V_1$, $\hat{\boldsymbol{u}} \in V$. Moreover, for any $\boldsymbol{v} = \boldsymbol{v}_0 + \boldsymbol{v}_1 \in V$:

$$\begin{aligned}\langle \boldsymbol{u} - \hat{\boldsymbol{u}}, \boldsymbol{v}\rangle &= \langle \boldsymbol{u} - \hat{\boldsymbol{u}}_0 - \hat{\boldsymbol{u}}_1, \boldsymbol{v}_0 + \boldsymbol{v}_1\rangle \\ &= \langle \boldsymbol{u} - \hat{\boldsymbol{u}}_0, \boldsymbol{v}_0\rangle - \langle \hat{\boldsymbol{u}}_1, \boldsymbol{v}_0\rangle + \langle \boldsymbol{u} - \hat{\boldsymbol{u}}_1, \boldsymbol{v}_1\rangle - \langle \hat{\boldsymbol{u}}_0, \boldsymbol{v}_1\rangle = \boldsymbol{0}.\end{aligned}$$

Since $\hat{\boldsymbol{u}} \in V$ and $\boldsymbol{u} - \hat{\boldsymbol{u}} \in V^{\perp}$, $\hat{\boldsymbol{u}}$ is the orthogonal projection onto $V$. ■

**Proposition 5.2.** Suppose $\boldsymbol{P}$ is the matrix for orthogonal projection onto a closed linear subspace $V$. $\boldsymbol{P}$ is **self-adjoint**:

$$\langle \boldsymbol{P}\boldsymbol{u}_1, \boldsymbol{u}_2\rangle = \langle \boldsymbol{P}\boldsymbol{u}_1, \boldsymbol{P}\boldsymbol{u}_2\rangle = \langle \boldsymbol{u}_1, \boldsymbol{P}\boldsymbol{u}_2\rangle.$$

♦

**Proof.** Since $\boldsymbol{u}_2 = \boldsymbol{P}\boldsymbol{u}_2 + (\boldsymbol{I} - \boldsymbol{P})\boldsymbol{u}_2$ with $\boldsymbol{P}\boldsymbol{u}_2 = \hat{\boldsymbol{u}}_2 \in V$ and $(\boldsymbol{I} - \boldsymbol{P})\boldsymbol{u}_2 = \boldsymbol{u}_2 - \hat{\boldsymbol{u}}_2 \in V^{\perp}$,

$$\langle \boldsymbol{P}\boldsymbol{u}_1, \boldsymbol{u}_2\rangle = \langle \boldsymbol{P}\boldsymbol{u}_1, \boldsymbol{P}\boldsymbol{u}_2 + (\boldsymbol{I} - \boldsymbol{P})\boldsymbol{u}_2\rangle = \langle \boldsymbol{P}\boldsymbol{u}_1, \boldsymbol{P}\boldsymbol{u}_2\rangle.$$

Writing $\boldsymbol{u}_1 = \boldsymbol{P}\boldsymbol{u}_1 + (\boldsymbol{I} - \boldsymbol{P})\boldsymbol{u}_1$ likewise gives:

$$\langle \boldsymbol{u}_1, \boldsymbol{P}\boldsymbol{u}_2\rangle = \langle \boldsymbol{P}\boldsymbol{u}_1 + (\boldsymbol{I} - \boldsymbol{P})\boldsymbol{u}_1, \boldsymbol{P}\boldsymbol{u}_2\rangle = \langle \boldsymbol{P}\boldsymbol{u}_1, \boldsymbol{P}\boldsymbol{u}_2\rangle.$$

■

**Corollary 5.2.** For Euclidean space:

$$\boldsymbol{u}_1'\boldsymbol{P}'\boldsymbol{u}_2 = \langle \boldsymbol{P}\boldsymbol{u}_1, \boldsymbol{u}_2\rangle = \langle \boldsymbol{u}_1, \boldsymbol{P}\boldsymbol{u}_2\rangle = \boldsymbol{u}_1'\boldsymbol{P}\boldsymbol{u}_2.$$

That is, $\boldsymbol{P}$ is symmetric. ♣

## 5.1 Euclidean Space

**Proposition 5.3.** Suppose $V_0 \subseteq V$ are closed linear subspaces of Euclidean space. Let $\boldsymbol{P}_0$ and $\boldsymbol{P}$ denote orthogonal projection onto $V_0$ and $V$, then:

$$\boldsymbol{P}\boldsymbol{P}_0 = \boldsymbol{P}_0 = \boldsymbol{P}_0\boldsymbol{P}.$$

That is, projection onto $V_0$ is equivalent to projection onto $V$, followed by projection onto $V_0 \subseteq V$. The operator $\boldsymbol{P}_0\boldsymbol{P}$ is an *iterated projection.* ♦

**Proof.** For any $\boldsymbol{v} \in \mathcal{H}$, since $\boldsymbol{P}_0\boldsymbol{v} \in V_0 \subseteq V$, $\boldsymbol{P}(\boldsymbol{P}_0\boldsymbol{v}) = \boldsymbol{P}_0\boldsymbol{v}$. Thus, $\boldsymbol{P}\boldsymbol{P}_0 = \boldsymbol{P}_0$. Further, since $\boldsymbol{P}_0$ and $\boldsymbol{P}$ are symmetric in Euclidean space:

$$\boldsymbol{P}_0 = \boldsymbol{P}_0' = (\boldsymbol{P}\boldsymbol{P}_0)' = \boldsymbol{P}_0'\boldsymbol{P}' = \boldsymbol{P}_0\boldsymbol{P}.$$

■

**Proposition 5.4.** Suppose $V_0 \subseteq V$ are closed linear subspaces of Euclidean space. Let $\boldsymbol{P}_0$ and $\boldsymbol{P}$ denote orthogonal projection onto $V_0$ and $V$. Define $V_1$ as the subspace of $V$ that is orthogonal to $V_0$, $V_1 = V_0^{\perp} \cap V$. Then:

i. $V = V_0 \oplus V_1$ with $V_0 \perp V_1$.

ii. $\boldsymbol{P}_1 = \boldsymbol{P} - \boldsymbol{P}_0$, where $\boldsymbol{P}_1$ is orthogonal projection onto $V_1$.

♦

**Proof.** (i.) Suppose $\boldsymbol{v} \in V$, and let $\hat{\boldsymbol{v}}_0 = \boldsymbol{P}_0\boldsymbol{v}$. Since $\boldsymbol{v} - \hat{\boldsymbol{v}}_0 \in V$ and $\boldsymbol{v} - \hat{\boldsymbol{v}}_0 \in V_0^{\perp}$, $\boldsymbol{v} - \hat{\boldsymbol{v}}_0 \in V_1$. Writing $\boldsymbol{v} = \hat{\boldsymbol{v}}_0 + (\boldsymbol{v} - \hat{\boldsymbol{v}}_0)$ demonstrates $\boldsymbol{v} \in V_0 + V_1$. Conversely, since $V_0 \subseteq V$ and $V_1 \subseteq V$, $V_0 + V_1 \subseteq V$. Moreover, since $V_1 = V_0^{\perp} \cap V \subseteq V_0^{\perp}$, $V_1 \perp V_0$, and the sum $V = V_0 \oplus V_1$ is direct.

(ii.) Since $V$ is the direct sum of two orthogonal subspaces, the orthogonal projection onto $V$ is expressible as $\boldsymbol{P} = \boldsymbol{P}_0 + \boldsymbol{P}_1$. ■

**Example 5.1.** Suppose $\mathbb{E}^n$ is Euclidean space and $V \subset \mathbb{E}^n$ is a linear subspace, then $\mathbb{E}^n$ is expressible as $V \oplus V^{\perp}$. Every $\boldsymbol{u} \in \mathbb{E}^n$ is uniquely expressible as $\hat{\boldsymbol{u}} + \boldsymbol{u}_{\perp}$, where $\hat{\boldsymbol{u}} = \boldsymbol{P}\boldsymbol{u}$ is the orthogonal projection onto $V$, and $\boldsymbol{u}_{\perp} = (\boldsymbol{I} - \boldsymbol{P})\boldsymbol{u}$ is the orthogonal projection onto $V^{\perp}$. The orthogonal projections are guaranteed to exist by closure of $V$ and $V^{\perp}$. Moreover, the two projections are orthogonal $\langle \hat{\boldsymbol{u}}, \boldsymbol{u}_{\perp} \rangle = 0$. ♠